Different Methods of Insect Pest Control

Pest control involves preventing insects from migrating, stopping their population growth, and destroying existing large populations. Because insects multiply rapidly in favorable conditions, control methods are mainly classified into two categories:

A. Natural Control

This is the automatic regulation of insect populations by natural, seasonal, physical, and biological factors.

***Seasonal Changes*:** Elements like temperature, rainfall, humidity, light, cold, heat, wind speed/pressure, and snowfall naturally affect pest numbers.

Topography (Land Structure): Geographical features such as mountains, deserts, swamps, forests, rivers, lakes, oceans, and soil structure greatly impact how insects spread.

Natural Enemies: Many destructive organisms survive by feeding on insects. This includes parasitoids, predatory insects, parasitic nematodes, frogs, lizards, bears, insectivorous animals, and birds (like mynas, quails, crows, and sparrows). Mutual conflict and cannibalism among insects are also important natural control factors.

***Natural Diseases*:** Insects naturally suffer from diseases caused by protozoa, bacteria, fungi, and viruses. These can cause epidemics that naturally wipe out pest populations.

B. Artificial Control

These are control measures applied by humans using various tools and techniques when natural factors fail to keep harmful insects in check.

Pest Biology: Complete understanding of the pest's life cycle, seasonal history, behavior, habitat, the type of damage it causes, and its natural enemies.

Identification: Accurate knowledge of the insect's classification and morphology (physical structure).

Method Selection: Determining the most effective techniques from the available control methods.

Economic Parameters: Determining the "Economic Threshold" and "Economic Injury Level" to decide exactly when control actions should be initiated.

1. Physical Control

This method utilizes physical elements to destroy pests, such as temperature (heat/cold), humidity, light, sound, electricity (energy), and radiation.

***Temperature*:** The optimal temperature for insect growth is 10–35°C.

1.High Heat: Exposing insects to 60–66°C for a short period kills most of them. Sun-drying infested grain at 52–55°C for 3–4 hours eliminates storage pests.

2.Cold: Storing fruits and vegetables at low temperatures protects them from pests.

Light: Light sources are used to attract and kill nocturnal insects (e.g., tobacco caterpillars, white grubs).

Note: Knipling (1937) successfully used gamma radiation to control the screwworm fly in America.

Humidity/Moisture: Manipulating moisture impacts pests. Draining stagnant water or swamps destroys mosquito and fly habitats. For safe grain storage, moisture levels should be kept below 10% (ideally 8–10%) to prevent most pest outbreaks (except the Khapra beetle).

***Sound*:** High-frequency waves are being used in places like Canada for pest control. Electronic flashguns are being developed to sterilize male insects.

***Electricity*:** Ultraviolet (UV) light traps are commercially available to attract and electrocute/kill flies, mosquitoes, and moths.

2. Mechanical Control

This method involves using human rationale, manual labor, or hand-operated tools to directly destroy pests before they reach the crops. While ancient, expensive, and time-consuming, it includes the following practical methods:

***Hand-picking and Destroying*:** Manually collecting and destroying egg masses, larvae, adult insects, or infested plant parts. This is highly effective in limited areas against localized or gregarious insects (e.g., hairy caterpillars, cabbage butterflies, lemon butterflies).

***Using Hand Nets*:** Catching flying or active insects (like grasshoppers, sugarcane pyrilla, and citrus butterflies) with hand nets to reduce their numbers.

Fencing: Installing physical barriers or fences around fields to stop crawling insects from migrating between fields.

***Trenching*:** Digging a trench (30–60 cm wide and 60 cm deep) around the field. Marching pests like locust hoppers or hairy caterpillars fall into these trenches during migration and are destroyed.

***Tree Banding*:** Tying a 20–30 cm wide slippery band (like an alkathene sheet) or applying a sticky band (using grease) around tree trunks (about 1 foot above the ground). This prevents flightless, crawling pests, like the mango mealybug, from climbing up the tree.

Traps

Traps are used to attract, collect, and destroy various types of insects.

(A) Light Trap: Used to attract nocturnal and light-loving insects.

- **Setup:** Uses a light source (mercury vapor bulbs are most effective, or UV tubes/lanterns) placed over a wide vessel containing water, kerosene oil (5:95 ratio), and insecticide.
- **Usage:** Best operated during peak emergence hours (7:30 PM to 10:30 PM).

    - **Effectiveness:** Kills moths (like those of white grubs, hairy caterpillars, cutworms) before they lay eggs. It is a simple, cheap, and highly effective method, especially during the rainy season.
- **(B) Sticky Trap:** Made by coating cardboard, gunny bags, wood, or paper with glue or sticky substances mixed with insecticide.
    - **Usage:** Hung on trees or bamboo sticks around the field.
    - **Effectiveness:** Small insects like aphids, jassids, and thrips get stuck and die. Flypaper is a great example (used to prevent apple maggots).
- **(C) Pheromone Trap:**Uses synthetic sex pheromones (chemicals naturally secreted by insects to attract mates). These emit a specific scent that lures target pests into the trap.
    - **Advantages:** Highly effective and safe for beneficial parasites and predators. Uses a rubber "Lure" specific to the pest (e.g., for gram, tomato, or cotton borers).
    - **Usage:** Generally applied at a rate of 5 traps per hectare, spaced 50 meters apart. The trapped insect count helps farmers decide whether chemical pesticides are needed.

Other Mechanical Activities

- **Wire Probing:** Destroying bark-eating caterpillars hiding in the trunks of fruit trees (lemon, guava, ber, pomegranate) by inserting and twisting a sharp wire into their tunnels.
- **Fruit Bagging:** Covering fruits (like pomegranate and grapes) with paper or cloth bags to protect them from pests.
- **Burning:** Using torches or flamethrowers at night to burn locust swarms.

3. Cultural Control

This method involves making timely adjustments to regular agricultural (agronomic) practices to reduce or eliminate pest populations. It utilizes standard crop production and management techniques to keep pests at bay.

- It is the simplest, cheapest, most practical, and longest-lasting method. It is also much safer for natural enemies compared to chemical pesticides.

Key Cultural Control Practices:

1.Post-Harvest Ploughing and Weeding/Hoeing: Soil structure, moisture, and temperature heavily influence a pest's life cycle. Deep ploughing after harvest brings hidden life stages (eggs, grubs, pupae) out of the soil. Once exposed, they are either eaten by natural enemies (like birds) or die due to adverse weather (harsh sunlight or severe cold).This exposure kills underground pests, cutworms, hairy caterpillars, grasshoppers, fruit flies, mango mealybugs, and red pumpkin beetles. Ploughing field boundaries before the monsoon easily destroys locust egg pods. Ploughing should be timed to avoid harming natural enemies or depleting soil moisture. Removing weeds also drastically reduces pest numbers.

- ***Adopting Crop Rotation*:** Planting crops of different botanical families in succession on the same land without depleting soil fertility. Since pests specific to one plant family rarely attack a different family, this breaks their life cycle.
    - *Example:* Sowing non-leguminous crops alongside groundnut reduces the outbreak of the leaf miner insect.
- ***Altering Sowing, Planting, and Harvesting Times:*** Every pest has a specific peak outbreak period. Adjusting agricultural timings helps the crop avoid the pest's most damaging stage.
    - *Examples:* Early sowing of mustard and cotton reduces aphid and jassid attacks. Early sowing of wheat reduces termite damage. Sowing sorghum in the first week of July effectively controls the shoot fly. Planting potato seeds slightly deeper prevents the potato tuber moth from laying eggs. Harvesting immediately upon maturity saves crops from borers and weevils (e.g., sugarcane borer, sweet potato weevil).
- ***Use of Resistant Varieties*:** Planting pest-resistant crop varieties is a highly successful, environment-friendly method. Host plant resistance is an inherited genetic trait that allows a plant to avoid, tolerate, or recover from pest damage. It is classified into three groups:
    1. **Tolerance:** The plant survives and yields despite pest attack.
    2. **Antibiosis:** The plant has biochemical traits that negatively affect the insect's life cycle, behavior, or metabolism.
    3. **Antixenosis (Non-preference):** The plant's physical or chemical traits make it unattractive to the pest for feeding or egg-laying.
- ***Clean Cultivation (Sanitation):*** Crop residues (stubble, stalks) left in the field act as breeding and hiding grounds for pests (especially for paddy, sugarcane, cotton, sorghum, maize, and brinjal). Destroying these residues kills resting borer larvae. Removing alternate host plants and weeds eliminates the pest's shelter.
- ***Balanced Fertilizer and Irrigation Management:*** Raw cow dung attracts termites, so only fully decomposed (rotted) manure should be used. Imbalanced fertilizers (like excess nitrogen) can trigger outbreaks of pests like sugarcane pyrilla, while low nitrogen can increase whiteflies. Applying a balanced dose of nutrients (NPK and micronutrients) boosts crop health without multiplying pests.
    - **Irrigation:** Timely irrigation keeps crops healthy and reduces pest/disease incidence. However, *excessive* irrigation and over-fertilization can actually increase pest and disease outbreaks.
- ***Cultivation of Mixed Cropping:*** Sowing different types of crops together increases the physical distance between plants of the same species. The secondary crop acts as a physical or biological barrier, restricting pest movement.
    - *Examples:* Mixed sowing of wheat, gram, and mustard reduces termite, pod borer, and aphid attacks, respectively. Sowing sorghum and pearl millet (bajra) together reduces the outbreak of the sorghum stem borer.

***Trap Cropping*:** Planting a specific crop (either as a prior or alternate crop) to attract pests away from the main crop. The concentrated pests are then destroyed using pesticides. It usually covers about 5% of the field, protecting natural enemies elsewhere.

- *Examples:* Sowing mustard 15 days before cabbage/cauliflower attracts the primary pests to the mustard instead of the main crop. Planting rows of maize around pumpkin and melon fields diverts fruit flies toward the maize.

Pruning and Thinning: Dense plant growth encourages pest outbreaks.

- *Pruning:* Old, densely planted mango orchards suffer heavily from Mango Jassids. Pruning branches allows light and air, reducing pest incidence and keeping fruits healthy.
- *Thinning:* Densely sown crops (like sorghum) attract more shoot borers. Thinning them out (maintaining proper spacing) keeps plants healthier and thicker, reducing pest attacks.

4. Chemical Control

Chemical control involves killing, repelling, or reducing pest populations using chemical substances. A "pest" is any organism whose population crosses a certain threshold, causing economic damage and conflicting with human welfare and agricultural profit.

Pesticides : Chemicals used to kill or repel organisms that harm humans, agriculture, or livestock. They are classified based on the target organism:

1. **Insecticides:** Effective against insects.
2. **Miticides (Acaricides):** Kill mites.
3. **Nematicides:** Destroy nematodes (roundworms).
4. **Rodenticides:** Kill rodents (e.g., rats).
5. **Molluscicides:** Kill snails and slugs.
6. **Fungicides:** Prevent/control fungi.
7. **Bactericides:** Prevent/control bacteria.
8. **Herbicides:** Destroy unwanted herbs/plants.
9. **Weedicides:** Destroy weeds.

1. Insecticide Chemicals

These are toxic chemicals specifically designed to kill or repel insect pests to protect crops, trees, and stored grains.

- **Characteristics:** Highly effective with a rapid **"Knockdown Effect."** They can quickly reduce large pest populations to a minimum or eradicate them.
- **Usage:** Due to being easy, successful, relatively cheap, and fast-acting, this is the most popular crop protection method. However, it should ideally be used only in special circumstances or emergencies.
- **Brief History:** Sulfur was used as an insecticide 100 years BC, and arsenic since 40 AD. The discovery of DDT in 1939 during WWII sparked a massive revolution in the development of synthetic insecticides.

2. Insecticide Formulation

Raw active chemicals are modified into "formulations" to make them easy and practical to apply in the field. Choosing the right formulation is critical for success. They are mainly available in two forms: Solid and Liquid.

Solid Formulations:

These are of three types

***1.Insecticidal Dust*:** Contains an active ingredient concentration ranging from 0.1% to 25%. In the market, it is commonly available in powder form at 2%, 5%, and 10% concentrations

Application of insecticidal dust is most beneficial during the early morning hours.

2. Insecticide Granules

- Prepared by combining the active insecticide with inert materials.
- Granule size ranges from 0.25 to 2.38 mm. They contain 2% to 10% active ingredient.
- Applied directly by broadcasting onto the soil or plants. No water is needed.

3. Wettable Powder (WP)

- Since most organic insecticides are insoluble in water and settle quickly, an inert wetting agent is added to keep the particles suspended in solution.
- Usually contains 25%, 40%, or 50% active ingredient.
- Effectiveness can be boosted by adding 1–2% surface-active agents (wetting or spreading agents).

Liquid Formulations

1. Emulsifiable Concentrate (EC)

- A common formulation made by mixing the insecticide, a solvent, and an emulsifier.
- The emulsifier provides stability and helps the liquid wet and spread across surfaces.
- When mixed with water, it forms an emulsion (like oil droplets suspended in water). It leaves a toxic residue on treated crops to kill pests.

2. Solution

- Synthetic organic insecticides that are insoluble in water are dissolved in organic solvents (like oil, ether, or paraffin).
- Used directly "as is" via hand sprayers to control household pests, mosquitoes, and aquatic insects (e.g., Flit spray).

3. Concentrate Liquid Insecticide

- A special liquid formulation that is **not** diluted with water.
- Used in Ultra Low Volume (ULV) spraying.
- *Examples:* Phosphamidon dissolved in isopropyl alcohol, or 95% Malathion concentrate.

4. Aerosol

- Consists of microscopic insecticide particles suspended in the air like mist or fog.

- High-pressure gas forcefully releases these particles, which slowly settle on crops. Pests die after eating the toxic leaves.
- Often used for large-scale aerial spraying (via airplanes or helicopters) during massive outbreaks, like locust swarms.

5. Fumigant

- Insecticides applied in their volatile (gaseous) state. They are stored as pressurized liquids.
- Used in enclosed spaces like grain storage bins, silos, godowns, and ship holds (e.g., Aluminium Phosphide tablets).

3. Classification of Insecticides

Insecticides can be fundamentally classified into three categories based on:

1. **Mode of Entry** (how they enter the insect's body)
2. **Mode of Action** (how they kill the insect)
3. **Nature of Chemicals / Active Ingredients**

(1) Based on Mode of Entry

***1. Contact Poison*:** These show their lethal effect simply by coming into contact with the insect's outer shell (cuticle). They are absorbed into the organs and kill the pest.

- Applied as dusts or sprays.

2. Stomach Poison: These are applied directly to the insect's food source (leaves, stems). When insects with chewing or sucking mouthparts ingest the treated plant parts, they die due to the chemical's toxicity.

3. Fumigant

- These volatile insecticides release poisonous gases.
- They enter the insect's respiratory tubes through their spiracles (breathing pores), causing asphyxiation (suffocation) and death.
- Used in airtight/sealed spaces primarily to control pests in stored grains.

(2) Based on Mode of Action

Insecticides can be classified by how they affect specific physiological systems once inside the insect's body:

1. ***Physical Poison:*** These are not true biological poisons. They kill insects mechanically by causing asphyxiation (blocking airways) or severe dehydration.
2. ***Protoplasmic Poison:*** Absorbed along with digested food in the gut, these poisons affect the proteins within the cell's protoplasm. They cause protein precipitation, leading to cell death and ultimately the insect's death.
3. ***Respiratory Poison:*** These enter the respiratory system and deactivate essential respiratory enzymes, killing the insect.
4. ***Nerve Poison:*** These enter the body and attack the nervous system. They dissolve in tissue fat and inhibit the *acetylcholinesterase* enzyme, causing the insect to become paralyzed (faint) and die.

(3) Based on Chemical Nature

Insecticides are primarily divided into two main categories based on their chemical makeup: Inorganic and Organic.

1. Inorganic Insecticides

- Usage of these is almost negligible today. They were heavily used before the advent of synthetic organic compounds.
- They are usually mineral or metal-derived and generally act as stomach poisons. Main types include:
    - **(i) Arsenical Compounds:** Highly toxic internal poisons for both insects and mammals. *Examples:* Lead arsenate, Calcium arsenate, Sodium arsenate, Paris green (Copper aceto-arsenite), and Arsenic oxides.
    - **(ii) Fluorine Compounds:** Less toxic to mammals compared to arsenic, highly toxic to insects, and safe for plants. They act as both contact and stomach poisons. *Examples:* Sodium fluoride, Sodium fluoaluminate, and Cryolite.
    - **(iii) Sulphur and its Compounds:** Mainly used as miticides (against mites) and fungicides. They have a low residual effect but can blacken metals, fade clothes, negatively affect seed germination, and emit a foul odor. *Examples:* Sulphur dust, Wettable sulphur.
    - **(iv) Phosphorus Compounds:** The most common is Zinc Phosphide. While historically used against grasshoppers and crickets, it is now primarily used as a poisonous bait for rodent (rat) control.

2. *Organic Insecticides* This is a large group that includes several sub-categories:

- **(i) Animal Insecticides and their Derivatives:** * *Example:* Nereistoxin (commercial name: Padan).
    - Derived from marine annelids (*Lumbriconereis heteropoda* and *L. brevicirra*). It kills insects by paralyzing their central nervous system.
- **(ii) Botanical (Plant) Insecticides and their Derivatives:** * Extracted from plant parts (roots, stems, leaves, flowers, and seeds).
    - **Benefits:** Completely safe for plants (phytotoxic-free) and have very low toxicity to mammals. Usually applied as contact poisons.
    - *Examples:* Nicotine, Pyrethrum, and Rotenone.
    - **Pyrethrum specific note:** Extracted by crushing the flowers of *Chrysanthemum cinerariifolium* and *Chrysanthemum coccineum*. It breaks down rapidly into non-toxic substances when exposed to sunlight, air, and moisture.To maintain its pest-killing properties, Pyrethrum must be stored in colored bottles or sealed containers in a cool, dry place. It is available as dust, wettable powder, and aerosol formulations.
- **Neem:** Native to India (Scientific name: *Azadirachta indica*). While all parts have insecticidal properties, the seeds (nimboli) are the most potent. Its primary active ingredient is **Azadirachtin**. It acts as a highly effective repellent, anti-feedant, and

insecticide against pests like desert locusts, brown plant hoppers, rice leaf folders, armyworms, diamondback moths, pod borers, and bollworms.

- **(iii) Oils and their Emulsions:** These are complex mixtures of hydrocarbons, primarily petroleum oils and coal tar oils. They are used in limited quantities on crops to avoid phytotoxicity (plant damage). Applied as emulsions at a safe concentration of 1% to 3%. They kill pests by asphyxiation (suffocation) and are effective against scale insects, mealybugs, whiteflies, mites, and their eggs.
- **(iv) Organic Thiocyanates:** Toxic to eggs, larvae, and adults. These are contact poisons that paralyze and kill flying insects. *Common examples:* Thanite, Lethane 60, and Lethane 384.
- **(v) Sulphonate, Sulphide, and Sulphon:** These compounds act primarily as contact miticides (killing mites and their eggs/nymphs). *Examples:* Tetradifon, Aramite, Ovex. However, their use is now quite limited due to plant toxicity issues and mites developing resistance.
- **(vi) Dinitrophenol:** Effective against eggs, larvae, and adults of both insects and mites through contact and stomach poisoning. Like sulphonates, their agricultural use is highly limited because they are toxic to plants.
- **(vii) Chlorinated Hydrocarbons (Organo-chlorine compounds):** Developed after World War II, these are considered the first "modern" synthetic insecticides. The use of DDT and BHC brought a revolutionary change in pest control.
    - **Properties:** Contain chlorine, hydrogen, and carbon. They are highly effective contact and stomach poisons, especially against insects with biting and chewing mouthparts.
    - **Regulatory Status:** Due to severe environmental persistence, many are heavily regulated today.
        - **Banned:** Endrin, Chlordane, Heptachlor, Aldrin, Toxaphene.
        - **Restricted Use:** BHC, Lindane, DDT, Dieldrin.
        - *Other Examples:* Endosulfan.
- **(viii) Organo-phosphorus Insecticides:** These phosphorus-containing chemicals are more capable and effective than organochlorines. They act as nerve poisons by inhibiting the *acetylcholinesterase* enzyme.
    - **Characteristics:** Pests have not yet developed high resistance to them. However, they are **extremely toxic to mammals**, requiring strict safety precautions during application. Highly effective against sap-sucking pests (aphids, jassids, whiteflies, thrips, mealybugs) and mites.
    - **Classification by Mode of Action:**
        - **(a) Contact and Stomach Poisons:** These remain active only on the treated surface and are not absorbed into the plant. *Examples:* Malathion, Methyl parathion, Phosalone, Chlorpyrifos, Quinalphos, Profenofos, Ethion.
        - **(b) Systemic Poisons:** These are absorbed by the plant (via roots or leaves) and spread throughout its internal system (sap). When pests feed on any part of the plant, they ingest the toxin and die. *Examples:* Phorate, Dimethoate, Methyl demeton, Phosphamidon, Monocrotophos, Acephate.
- **(ix) Carbamate Insecticides:** These are derivatives of organic carbamic acid. Some also possess molluscicidal properties (killing snails). They act as contact, stomach, and systemic poisons for insects.

Despite their toxicity to insects, they are considered very safe for mammals. *Examples:* Carbaryl and Carbofuran.

- **(x) Fumigants:** These are substances that vaporize into toxic gases at ordinary temperatures.
    - **Action:** The gas enters the insect's respiratory tubes (trachea) through their spiracles, attacks the nervous system, and kills them.
    - **Usage:** Applied in enclosed/airtight spaces like godowns, mills, ship holds, storage bins, or under airtight tarpaulin tents. They penetrate rapidly (especially in a vacuum) and reach deep into cracks, crevices, and between grain bags to kill hidden storage pests that contact or stomach poisons cannot reach. Also used effectively against nematodes and rodents.
    - **Examples:** Hydrogen cyanide, Methyl bromide, Chloropicrin, Paradichlorobenzene, Naphthalene, Ethylene dichloride, EDCT, Ethylene dibromide, DD mixture, and Aluminium phosphide.
    - *Caution:* Many of these are highly toxic, restricted, or banned, and must only be used by trained personnel with extreme caution.
- **(xi) Synthetic Pyrethroids:** Highly active against insects, exceptionally low toxicity to mammals, stable, effective at very low doses, fast-acting, and leave harmless degradable residues. They act primarily as contact poisons and somewhat as stomach poisons.
    - **Usage:** Safely used for agricultural crops, stored grain pests, household pests, and parasites on pets and livestock. They easily control stubborn pests like stem borers and bollworms where other strong insecticides fail.
    - *Examples used in India:* Fenvalerate, Cypermethrin, Deltamethrin, Lambda-cyhalothrin.

4. Miticides (Acaricides)

These are specific chemicals designed to kill plant-feeding (phytophagous) mites.

- Mites suck the sap from plant leaves, buds, and flowers, causing severe weakness and spreading various diseases. Because they are microscopic, they often go unnoticed until massive populations cause heavy damage.

- **Chemical Control History & Examples:** * Sulphur was first used for mite control in 1920 and continues to be used today.
    - Later, specific chlorinated hydrocarbons and new synthetic miticides were developed.
    - *Examples:* Dicofol, Chlorobenzilate, and Carbophenothion.

5. Rodenticides

These are chemicals used to kill rodents like rats, mice, and squirrels. Rodents are highly clever, suspicious, rapid-breeding pests found almost everywhere, causing massive damage to crops, stored grains, and household items. Rodenticides are classified into three categories:

(A) Acute Rodenticides:

- **Action:** Single-dose poisons. Rodents die within a few hours of consuming just one dose.
- **Examples:** Zinc phosphide, Barium carbonate, ANTU, and Thallium sulfate.
- **Application (Zinc Phosphide Bait):** The most commonly used. A poison bait is prepared by mixing 2 parts Zinc phosphide + 97 parts crushed grain (bajra, jowar, maize, or wheat) + 1 part edible oil. This is placed at night on small pieces of paper or clay pots near burrows and pathways.
- **Drawback & Solution ("Bait Shyness"):** Rats can detect the foul smell of the poison and become suspicious, refusing to eat it. To trick them, **pre-baiting** is required: feed them plain, unpoisoned bait for 2-3 days to build their habit, then replace it with the poisoned bait for high success.

(B) Chronic Rodenticides:

- **Action:** Multiple-dose poisons, also known as **Anticoagulants**. The bait must be consumed over several days (about a week). They have a cumulative effect that prevents blood clotting, causing the rat to die from internal and external bleeding.
- **Advantage:** No pre-baiting is needed. Rats eat these baits eagerly without suspicion or fear.
- **Examples:** Warfarin (used at 0.025% active ingredient mixed with grain), Rodafarin, Coumafuryl, and Racumin.
- **Modern Anticoagulants:** Newer, highly popular options include Bromadiolone (Moosh-Moosh, Supercaid), Flocoumafen, Brodifacoum (Talon), and Chlorophacinone (Drrat).

(C) Fumigant Rodenticides:

- **Action:** Poisonous gases used to suffocate rodents.
- **Example:** Aluminium phosphide is a major fumigant, highly effective for directly treating and sealing rat burrows in the fields.

Biological Control

The successful control of pests by humans through the encouragement, introduction, and dissemination of other living organisms (parasitoids, predators, and pathogens).

- **Natural Enemies:** Include parasitic and predatory insects, disease-causing viruses, bacteria, fungi, protozoa, parasitic nematodes, and predatory vertebrates.
- Based on the concept of *"Jivo Jivasya Bhojanam"* (One living organism is food for another). Examples: Grasshopper-Myna, Snake-Mongoose, Rat-Cat.
- Practiced since ancient times. In 1762, the Indian Myna was sent to Mauritius to control red locusts.
- Ants were historically used to control pests in citrus and date palms.
- *Apanteles glomeratus* was used in Italy against the cabbage caterpillar.
- The Ladybird beetle (*Rodolia cardinalis*) was imported from Australia to California (USA) to successfully control the Cottony cushion scale (*Icerya purchasi*) on citrus plants.
- Systematic work began in England in 1927 at the Farnham House Laboratory, rearing the parasitoid *Aphelinus mali* for Apple woolly aphid control.

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Integrated Pest Management (IPM)

The coordination of all possible pest management methods into a single program that is economically, socially, and ecologically beneficial.

The discovery of organic pesticides (like DDT and BHC) around 1945 during WWII led to massive, blind usage for pest and mosquito control until the 1960s.

- The term "Integrated Pest Control" was first used by Bartlett in 1956 (combining chemical and biological controls).
- In 1970, Geier introduced the term **"Integrated Pest Management" (IPM)**.
- *Note: Dr. R.F. Smith and Dr. P.L. Adkisson received the World Food Prize in 1997 for their pioneering work in developing and implementing IPM concepts to reduce the use of deadly toxic pesticides.*

Core Principles of IPM:

1. **Economic Threshold Level (ETL):** The mere presence of an insect in a field does not mean it will cause economic loss. Pesticides should only be used when the pest population crosses a specific limit where they start causing economic damage. This limit is called the Economic Threshold Level.
2. **Ecological Balance:** Having a small number of harmful pests in the field is actually ecologically beneficial. They serve as food for our "farmer-friendly" insects (predators and parasitoids). Completely eradicating pests disrupts the entire food chain.
3. **Objective:** The goal of IPM is not complete eradication, but rather to maintain pest populations at a specific, manageable level while considering economic, social, and environmental values.

Equipments of Spraying and Dusting

1. Selection of Equipment

When choosing plant protection equipment, consider the following:

- It should be made of non-corrosive metal or plastic.
- Its capacity, design, and mechanism must suit your specific needs.
- Spare parts, repair, and maintenance facilities should be easily available.
- The price should be affordable for the farmer.
- It should require less manual labor and be highly durable.

2. Classification of Equipments

(A) Dusters :

1. **Hand-Operated:** Rotary, Plunger, Bellows, etc.
2. **Power-Operated:** Power duster, Mist blower.

(B) Sprayers :

1. **Hydraulic (Liquid-operated):** Uses a pump to pressurize the liquid. Includes Knapsack sprayers, Hand sprayers, and Stirrup pumps.
2. **Pneumatic / Air-compression:** Compression knapsack sprayers.
3. **Gaseous Power:** Hand atomizers.
4. **Power-Operated Sprayers:** For large areas (Hydropneumatic, Mist blowers, High-pressure hydraulic).
5. **Aerial Sprayers:** Airplanes and helicopters.

Key Equipment

Knapsack Sprayer

A manually operated, continuous-pressure hydraulic sprayer carried on the operator's back like a knapsack.

- **Construction:** The tank is flat or bean-shaped, made of galvanized iron, brass, stainless steel, or plastic. Capacity is usually 10–16 liters (can be up to 30L). Main parts include the tank, pump, and a lance with a nozzle.
- **Mechanism:** An agitator/pump inside the tank barrel is operated by an up-and-down movement of a hand lever. This forces the liquid past a foot valve into a pressure chamber, and finally out through the discharge hose and nozzle as a fine spray. The spray is controlled by a trigger/valve on the lance.
- **Usage:** Ideal for marginal/small farmers to spray low-height crops, nurseries, vegetables, and small trees (up to 2.5m). One person can cover more than 1 hectare per day. It operates easily at a pressure of 3–5 kg/cm² and is excellent for rough, hilly, or marshy terrains.

Hand Rotary Duster

Also known as a crank duster or fan duster. It is designed to be hung from the shoulder or strapped to the chest/back. It consists of three main parts:

1. **Gear Box:** Contains a series of gears. A large main gear is connected to a crank handle. Turning the crank slowly rotates the large gear, which in turn rotates smaller gears connected to a fan shaft at very high speeds.
2. **Hopper:** A drum-like container that holds 4–6 kg of insecticidal dust (filled to 3/4 capacity). It has a lid on top and a chest plate at the back. Inside, a slow-moving crank-shaft with bent rods (**Agitators**) constantly shakes the dust to prevent it from settling. A **Feeder** plate pushes the dust toward the exit, regulated by an external **Feed Regulator**.
3. **Blower:** A closed box containing a 5-blade fan. As it spins, it throws air forward, creating a vacuum that sucks the dust from the hopper. The dust-air mixture is pushed through a long rubber tube (**Lance**). The tip of the lance has a **Nozzle** with a **Deflector** (to direct the dust up or down). A Y-attachment can be used to dust two rows simultaneously.

- **Working Mechanism:** Strap the machine to your chest, attach the lances, and adjust the deflector (point down for upper leaves, point up for the underside of leaves). Turn the crank handle while walking.
- **Efficiency:** Highly useful for low-height crops, vegetables, small bushes, home gardens, and godowns. Covers 1 to 1.5 hectares per day.

सोचने से कहाँ मिलते है तमन्नाओ के शहर,
चलना भी जरुरी है मंजिल पाने के लिए!